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The Resonant Human: High-Frequency Hemodynamics, Mechanotransduction, and the Pathogenesis of Arteriovenous Fistula Failure

1. Introduction: The Frequency Spectrum of Vascular Biology

The human body is a dynamic mechanical entity that exists in a perpetual state of oscillation. From the fundamental low-frequency periodicity of the cardiac cycle (1–2 Hz) to the kilohertz-range firing of neuronal action potentials, physiological function is intrinsically linked to frequency-dependent mechanical and electrical events. In the context of vascular biology, the prevailing paradigm has long focused on the low-frequency domain—specifically, the pulsatile forces of blood pressure (cyclic strain) and the frictional drag of blood flow (wall shear stress) generated by the beating heart. These forces, operating typically below 2 Hz, are the primary regulators of vascular homeostasis, dictating vessel diameter, wall thickness, and endothelial quiescence.

However, a rapidly emerging body of high-fidelity research suggests that this low-frequency view is incomplete. Pathological states, particularly those involving high-velocity blood flow such as the arteriovenous fistula (AVF) created for hemodialysis, introduce a new spectrum of mechanical energy: high-frequency vibrations (HFV). These vibrations, ranging from 20 Hz to over 1000 Hz, are not merely passive byproducts of turbulent flow; they are active biological stimuli. Recent investigations utilizing monolithic fluid-structure interaction (FSI) solvers, advanced phonoangiography, and molecular mechanobiology have revealed that vascular cells—much like the specialized mechanoreceptors in the skin or the hair cells of the cochlea—are sensitive to these supraphysiological frequencies.

This report provides an exhaustive synthesis of the interactions between high-frequency mechanical energy and human biological tissues, with a primary focus on the arteriovenous fistula as a critical clinical model. By integrating computational fluid dynamics, acoustic signal processing, and comparative mechanobiology, we elucidate how the "bruit" and "thrill" of an AVF drive the adverse vascular remodeling that leads to access failure. Furthermore, we explore the broader implications of frequency-dependent mechanotransduction, drawing parallels between vascular injury, bone remodeling, and sensory perception to present a

unified theory of the resonant human body.

1.1 The Hemodynamic Paradox of the AVF

The arteriovenous fistula represents a deliberate hemodynamic catastrophe surgically induced to sustain life. By connecting a high-pressure artery directly to a low-pressure vein, surgeons bypass the high-resistance capillary bed, creating a low-resistance circuit capable of delivering the massive blood flow volumes (often exceeding 500–1000 mL/min) required for efficient extracorporeal blood purification.¹ This procedure forces the venous segment, evolutionarily adapted to steady, low-velocity laminar flow, to accommodate arterial-like pressures and highly chaotic, high-velocity flow regimes.

Historically, the failure of these fistulae—characterized by neointimal hyperplasia (NIH) and subsequent stenosis—was attributed to aberrant Wall Shear Stress (WSS). The "oscillatory shear stress" theory posited that regions of flow recirculation, where WSS is low and directionally changing, trigger pro-inflammatory endothelial phenotypes. While valid in arterial bifurcations (atherosclerosis), this theory has proven inconsistent in AVFs. Stenotic lesions frequently develop in regions where WSS metrics alone cannot account for the severity of the remodeling. This discrepancy has driven the search for a "missing mechanical link," leading researchers to the hypothesis that the kinetic energy of turbulent fluctuations, manifesting as high-frequency wall vibrations, constitutes a specific and potent driver of stenosis.³

2. The Physics of Biological Frequency: A Comparative Analysis

To understand how a blood vessel "feels" a 500 Hz vibration, one must first appreciate the broad capacity of biological tissues to detect and respond to oscillatory mechanical energy. The human body is equipped with specialized systems designed to transduce vibrations into electrochemical signals, and comparing these systems reveals conserved mechanisms that may also be active, albeit pathologically, in the vascular wall.

2.1 Sensory Mechanoreceptors: The Tuned Antennae

The somatosensory system provides the clearest example of frequency-tuned mechanotransduction. The skin contains discrete populations of mechanoreceptors, each "tuned" to a specific bandwidth of mechanical oscillation.

- **Meissner Corpuscles (Flutter):** Located in the dermal papillae, these receptors are optimally sensitive to low-frequency vibrations in the **40–60 Hz** range. They are responsible for detecting "flutter" or slip—the sensation of an object moving across the skin. This frequency band overlaps significantly with the fundamental "shedding frequencies" of unstable vortices in transitional blood flow.⁶
- **Pacinian Corpuscles (Vibration):** Deeply embedded in the dermis and subcutaneous

tissue, Pacinian corpuscles are the body's high-frequency sensors. They exhibit peak sensitivity between **200 Hz and 300 Hz**, capable of detecting displacement amplitudes as small as 10 nanometers. Structurally, the corpuscle consists of a central nerve fiber surrounded by concentric fluid-filled lamellae (an onion-like structure) that act as a mechanical high-pass filter, allowing only transient, high-frequency signals to reach the axon.⁶

Crucially, the molecular machinery underpinning these receptors—specifically the Piezo1 and Piezo2 ion channels—is also expressed in non-neural tissues, including vascular endothelial cells and red blood cells.⁸ This suggests that while a vein lacks the "onion" structure of a Pacinian corpuscle, its constituent cells possess the fundamental molecular apparatus to detect high-frequency energy if the amplitude is sufficient.

2.2 The Cochlea and the "Half-Octave Shift"

The auditory system offers a critical analog for vibration-induced damage. In the cochlea, sound waves (airborne vibrations) are converted into fluid waves that deflect the stereocilia of hair cells. Noise-induced hearing loss provides a specific mechanistic insight known as the **"half-octave shift."** When the ear is exposed to an intense tone at a specific frequency, the maximal structural damage and resulting hearing loss occur not at the stimulation frequency, but approximately half an octave *above* it.⁹

This phenomenon, driven by the non-linear biomechanics of the basilar membrane, is relevant to vascular mechanics. It implies that the frequency of the hemodynamic "noise" (the flow instability) may not perfectly align with the frequency of the maximal wall damage. Instead, non-linear interactions within the viscoelastic vessel wall could shift the focal point of injury, complicating the direct correlation between flow simulations and clinical stenosis locations.

2.3 Bone and Muscle Resonance

Beyond sensation, vibration acts as a metabolic regulator in structural tissues.

- **Osteoblasts:** Bone cells exhibit distinct resonance frequencies. Finite element models predict the first ten natural frequencies of osteoblasts to lie within the **6.85–48.69 Hz** range, with higher modes extending to **200 Hz**. Exposing bone to vibrations within these bands (e.g., 30 Hz) can enhance osteogenesis, increasing bone density.¹⁰
- **Myocytes:** In skeletal muscle, high-frequency vibration (HFV) modulates the expression of atrophy-related genes. Frequencies in the range of **30–60 Hz** have been shown to downregulate myostatin and atrogin-1, promoting hypertrophy.¹¹

This comparative biology establishes a fundamental principle: human cells are not mechanically inert to high frequencies. They possess resonant structures (cytoskeleton, nucleus) and molecular sensors (ion channels) that respond to vibrations well above the 1 Hz cardiac baseline. In the context of the AVF, the vein—which is effectively a muscular tube—is

inadvertently subjected to a "workout" of continuous, high-intensity vibration, triggering hypertrophic pathways (neointimal hyperplasia) that are maladaptive rather than strengthening.

3. Hemodynamic Origins of High-Frequency Energy

The generator of these vibrations in an AVF is the blood flow itself. The juxtaposition of high arterial velocity and venous curvature creates a perfect storm for fluid instability.

3.1 Transitional Flow and Turbulence

Blood flow in a mature AVF typically operates at Reynolds numbers (Re) ranging from 1,000 to over 4,000. This places the flow squarely in the "transitional" regime—a chaotic state that fluctuates intermittently between laminar and turbulent flow. Unlike fully developed turbulence (which is isotropic and chaotic at all scales), transitional flow is characterized by the periodic formation and breakdown of coherent flow structures.

- **Dean Vortices:** As blood navigates the "swing segment" (the curved portion of the vein leading away from the artery), centrifugal forces drive high-momentum fluid toward the outer wall. This sets up a secondary flow pattern consisting of counter-rotating helical structures known as Dean vortices.
- **Vortex Shedding:** At high Re , these vortices become unstable. They oscillate and "shed" from the wall at specific frequencies. The Strouhal number (St) describes this relationship: $f = (St \cdot V) / D$, where f is frequency, V is velocity, and D is diameter. In the AVF environment, these shedding frequencies often fall into the **40–200 Hz** range.¹³

3.2 Pressure Fluctuations vs. Wall Vibration

It is critical to distinguish between the energy in the fluid and the motion of the wall.

- **Turbulent Kinetic Energy (TKE):** This measures the intensity of velocity fluctuations within the blood. High TKE indicates "chaotic" blood flow.
- **Wall Pressure Fluctuations:** The chaotic flow exerts a rapidly changing traction force on the vessel wall.
- **Wall Vibration:** The wall responds to these forces based on its own mechanical impedance. A calcified, stiff artery might reflect this energy, whereas a compliant, thin-walled vein will absorb it, resulting in physical displacement.

Research utilizing high-fidelity simulations indicates that while TKE is distributed throughout the lumen, the actual wall vibration is highly focal. It is predominantly localized to the **anastomosis floor** (where the arterial jet impacts) and the **inner curvature** of the vein (where flow separation occurs).⁴ These are the precise anatomical sites where stenosis is

most frequently observed clinically.

4. Computational Fluid-Structure Interaction: The "TurtleFSI" Approach

Investigating high-frequency vascular mechanics in vivo is fraught with difficulty. Doppler ultrasound lacks the temporal resolution to capture 500 Hz vibrations, and invasive sensors alter the flow they seek to measure. Consequently, the field has advanced primarily through *in silico* experimentation using Fluid-Structure Interaction (FSI) models.

4.1 The Challenge of the "Added Mass" Effect

Simulating blood flow in a vein is computationally distinct from simulating air flow over a wing. In aerodynamics, the fluid (air) is much lighter than the solid (metal), allowing for "partitioned" solvers that solve the fluid and structure equations sequentially. In hemodynamics, the density of blood ($\rho_f \approx 1060 \text{ kg/m}^3$) is nearly identical to that of the vascular wall ($\rho_s \approx 1050 \text{ kg/m}^3$).

This density parity creates the **"added mass effect."** When the vessel wall moves, it must accelerate a significant mass of blood along with it. In a partitioned solver, this strong coupling causes numerical instability—the solution for the interface position oscillates wildly and diverges.

4.2 The Monolithic Solution: TurtleFSI

To overcome this, researchers have adopted **monolithic** solvers, such as the open-source **TurtleFSI** (built on the FEniCS platform).¹ A monolithic solver assembles the governing equations for both the fluid (Navier-Stokes) and the solid (St. Venant-Kirchhoff or Neo-Hookean) into a single, massive matrix system that is solved simultaneously.

Mathematical Formulation:

The system solves for velocity ($\mathbf{v}$) and displacement ($\mathbf{u}$) in a unified Arbitrary Lagrangian-Eulerian (ALE) framework:

$$\begin{bmatrix} \mathcal{F}_{\text{fluid}} & \mathcal{C}_{\text{fs}} \\ \mathcal{S}_{\text{solid}} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{v} \\ \Delta \mathbf{u} \end{bmatrix} = \begin{bmatrix} \mathcal{R}_{\text{fluid}} \\ \mathcal{R}_{\text{solid}} \end{bmatrix}$$

Where $\mathcal{C}_{\text{fs}}$ and $\mathcal{C}_{\text{sf}}$ represent the strong coupling terms at the interface. Solving this system is computationally expensive, as the Jacobian matrix assembly (required for Newton-Raphson iterations) dominates the processing time, often exceeding the time required for the linear solver itself.¹⁵

4.3 Simulation Parameters for High-Frequency Resolution

Capturing a 500 Hz vibration requires rigorous temporal and spatial discretization strategies that far exceed standard cardiovascular CFD standards.

- **Temporal Resolution:** To resolve frequencies up to ~1000 Hz, the time step (Δt) must satisfy the Nyquist-Shannon sampling theorem ($f_{\text{sampling}} > 2 \cdot f_{\text{max}}$), but for numerical stability in ALE formulations, it must be even smaller. Studies typically employ Δt in the range of **1.0×10^{-4} s to 1.5×10^{-5} s**.¹⁷
- **Spatial Resolution:** The mesh must resolve the "Kolmogorov length scale"—the smallest scale of turbulent energy dissipation. AVF models frequently utilize meshes of **150,000 to over 3 million** elements. Quadratic Taylor-Hood elements (P2-P1) are used to provide second-order accuracy for velocity and pressure fields.¹⁹
- **Boundary Conditions:** To isolate flow-induced vibrations from the cardiac pulse, some studies employ a steady-state inflow velocity. While physiologically simplified, this approach removes the low-frequency "noise" of the heartbeat, allowing researchers to attribute any observed wall motion solely to the flow instability.²⁰

4.4 Key In Silico Findings

1. **Magnitude of Displacement:** The vessel wall in a turbulent AVF vibrates with amplitudes in the range of **microns** (e.g., 1–50 μm). While small compared to the millimeter-scale expansion of the pulse, the *acceleration* associated with these high-frequency movements is significant.¹³
2. **Frequency Content:** The vibration spectrum is broad but exhibits distinct peaks. In cerebral aneurysm models (a similar geometry), peaks were observed at **40–60 Hz** and **255–265 Hz**.¹³ The 40–60 Hz peak closely aligns with the sensitivity of Meissner corpuscles, while the 250 Hz peak aligns with Pacinian corpuscles and the most damaging frequencies for endothelial cells.

5. The Acoustic Signature: Advanced Signal Processing

The clinical manifestation of these high-frequency wall vibrations is sound. Physicians have long used the stethoscope to listen for the "bruit" of a fistula. A continuous, low-pitched machinery murmur indicates healthy laminar flow; a high-pitched, harsh whistle indicates stenosis. Modern engineering has transformed this subjective art into a quantitative science known as **phonoangiography**.

5.1 Spectral Analysis and the "Bruit"

The sound produced by an AVF is a direct acoustic emission of the wall vibration. By placing an electronic stethoscope or a piezoelectric sensor on the skin, researchers can record these sounds and apply Fast Fourier Transform (FFT) algorithms to generate a Power Spectral Density (PSD) profile.

5.1.1 Frequency Bands of Interest

- **Low Frequency (20–100 Hz):** Corresponds to the palpable "thrill." These frequencies are linked to the structural resonance of the vessel and surrounding tissue.²¹
- **Mid Frequency (100–700 Hz):** The primary domain of the "bruit." Energy in this band is generated by turbulent eddies and flow separation.²³
- **High Frequency (> 700 Hz):** Often associated with severe, critical stenosis where the flow jet velocity is extremely high.

5.2 The High-Low Peak Ratio (HLPR)

A significant advancement in acoustic surveillance is the development of the **High-Low Peak Ratio (HLPR)**, a metric designed to quantify the "pitch" shift associated with stenosis.²⁴

Definition:

The HLPR compares the peak energy in the high-frequency band to the low-frequency band.

- **Low Band:** Typically defined as **100–250 Hz**.
- **High Band:** Typically defined as **500–700 Hz**.
- **Calculation:** $HLPR = \frac{PeakAmp_{High}}{PeakAmp_{Low}}$

Clinical Thresholds:

Data from recent prospective studies²⁴ indicates:

- **HLPR < 0.35:** Associated with **laminar flow**, low turbulence, and minimal Oscillatory Shear Index (OSI). The vessel is stable.
- **HLPR > 0.35:** Indicates **complex flow** with recirculation and high OSI. This is a strong predictor of progressive venous remodeling, stenosis, or aneurysmal dilatation.
- **Correlation:** Longitudinal tracking shows that as HLPR increases, the brachial blood flow volume (BFV) tends to decrease (Slope = -0.001, $p < 0.01$), validating HLPR as a functional biomarker.²⁴

5.3 Deep Learning and Mel-Spectrograms

To handle the complexity of acoustic data, researchers are increasingly employing deep learning. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are trained on **Mel-spectrograms**—visual representations of sound that mimic human auditory perception.²⁶

- **Mel Scale:** This scale expands the representation of lower frequencies and compresses higher frequencies, aligning with how the ear (and potentially biological tissue) integrates frequency information.

- Performance:** These models can classify AVF sounds as "stenotic" or "patent" with high accuracy (AUC > 0.90), often outperforming traditional spectral features alone.²⁶

5.4 Radar-Based Wall Motion Sensing

Beyond sound, electromagnetic sensing offers a non-contact method to measure wall vibration directly. **Pulse Radar sensors** (operating in the millimeter-wave or microwave bands) detect the micro-displacements of the vessel wall through phase shifts in the reflected signal.²⁹

- Mechanism:** The radar emits a pulse sequence. The chest wall or arm movement modulates the phase of the echo.
- Vascular Wall Motion (VWM):** This system isolates the specific vibration signature of the fistula.
- Machine Learning Integration:** Support Vector Machine (SVM) classifiers trained on radar-derived harmonic ratios have demonstrated a specificity of 100% and sensitivity of 95.2% in detecting flow dysfunction (defined as flow < 750 mL/min).³¹ This technology represents a leap forward from the qualitative "thrill check," offering a digitized, continuous monitoring solution.

Table 1: Comparative Technologies for High-Frequency Hemodynamic Assessment

Technology	Modality	Frequency Range	Measured Variable	Primary Insight
Doppler Ultrasound	Acoustic (MHz)	0 – 10 kHz (Shift)	Blood Velocity	Flow Volume, Lumen Diameter
Phonoangiography	Acoustic (Audio)	20 – 1000 Hz	Sound Pressure	Turbulence Intensity, HLPR
Pulse Radar	EM Wave (GHz)	0 – 500 Hz (Motion)	Wall Displacement	Wall Vibration Amplitude
Electronic Stethoscope	Piezoelectric	20 – 2000 Hz	Surface Vibration	"Bruit" Characterization

6. Comparative Mechanobiology: How Cells "Hear" the Flow

The convergence of computational and acoustic data confirms that the AVF wall vibrates at frequencies between 40 Hz and 700 Hz. The critical biological question remains: **How do vascular cells transduce these frequencies into pathological remodeling?** Standard mechanobiology focuses on static stretch or 1 Hz pulsatility. To explain the response to 400 Hz, we must look to comparative biology.

6.1 The Endothelial and Smooth Muscle Response

In vitro experiments exposing vascular cells to high-frequency vibrations (HFV) reveal distinct, stress-related phenotypes that differ from those induced by laminar shear.

- **Vacuolization:** A hallmark of HFV exposure (e.g., 60 Hz to 250 Hz) is the formation of large cytoplasmic vacuoles in both Endothelial Cells (ECs) and Smooth Muscle Cells (SMCs). These vacuoles resemble those seen in osmotic stress or massive vasoconstriction, suggesting severe cytoskeletal disruption.¹¹
- **Cytoskeletal Disassembly:** The actin cytoskeleton is a tension-tensegrity structure tuned to specific frequencies. Vibrations near the cell's resonant frequency (predicted at 10–200 Hz for adherent cells¹⁰) may cause the actin filaments to depolymerize or detach from focal adhesions. This loss of structural integrity triggers a "repair" response—proliferation.
- **Gene Expression (MEG3):** Vibration at frequencies like 63 Hz and 250 Hz has been shown to downregulate **lncRNA MEG3** in endothelial cells.¹¹ *MEG3* is a long non-coding RNA that typically suppresses cell proliferation. Its suppression removes the "brake" on the cell cycle, promoting the neointimal hyperplasia observed in stenosis.

6.2 The Inflammation-Oxidation Axis

HFV acts as a pro-inflammatory stimulus. Unlike laminar shear stress, which promotes the release of nitric oxide (NO) and maintains a quiescent state, high-frequency agitation triggers:

- **Oxidative Stress:** Upregulation of Reactive Oxygen Species (ROS).
- **Cytokine Release:** Increased secretion of IL-6 and other pro-inflammatory markers.¹¹
- **YAP/TAZ Signaling:** This mechanosensitive pathway, usually regulated by substrate stiffness, is also activated by high-frequency dynamic loading. YAP nuclear translocation drives SMC phenotypic switching from a contractile (healthy) to a synthetic (proliferative) state.¹¹

6.3 The "Vascular Cochlea" Hypothesis

We can conceptualize the stenotic lesion as a form of "acoustic trauma" to the vessel wall.

- **Frequency Matching:** The most damaging frequencies observed in cell culture (60 Hz,

250 Hz) closely match the spectral peaks identified in FSI simulations of AVFs¹³ and the sensitivity ranges of cutaneous mechanoreceptors.⁶

- **Localized Damage:** Just as specific frequencies damage specific hair cells in the cochlea, specific flow instabilities damage specific regions of the vein. The "toe" of the anastomosis, subjected to the high-frequency "shedding" of the arterial jet, undergoes the most aggressive remodeling.
- **Fatigue Failure:** Unlike the cyclic strain of the pulse (approx. 36 million cycles/year), a 300 Hz vibration subjects the cell to **9.4 billion cycles/year**. This orders-of-magnitude difference suggests that high-cycle fatigue of the extracellular matrix (ECM) and cellular adhesions plays a significant role in the degradation of the vessel wall.

7. Pathological Remodeling: The Vibration-Induced Stenosis Cycle

Synthesizing the hemodynamic, acoustic, and biological data allows us to construct a unified model of AVF failure: **The Vibration-Induced Stenosis Cycle**.

1. **Hemodynamic Trigger:** The surgical creation of the AVF initiates high-velocity transitional flow.
2. **Energy Conversion:** Flow instability generates Turbulent Kinetic Energy (TKE), which couples to the compliant venous wall as High-Frequency Vibration (HFV) (40–700 Hz).
3. **Mechanotransduction:**
 - **Physical:** Cytoskeletal resonance causes actin disassembly and vacuolization.
 - **Molecular:** Downregulation of *MEG3*, activation of YAP/TAZ and NF-κB inflammatory pathways.
4. **Cellular Response:**
 - **SMCs:** Switch to synthetic phenotype, migrate to the intima, and proliferate (Neointimal Hyperplasia).
 - **ECs:** Enter a pro-inflammatory, adhesive state, recruiting leukocytes.
5. **Structural Stiffening (The "Fix"):** The body attempts to dampen the damaging vibrations by thickening the wall (hyperplasia).
6. **The Vicious Loop:**
 - Intimal thickening narrows the lumen (Stenosis).
 - Stenosis increases flow jet velocity.
 - Higher velocity increases turbulence intensity and shifts the vibration to *higher* frequencies and amplitudes (Rising HLPR).
 - The cycle accelerates until thrombosis occurs.

This model explains why stenosis is notoriously recurrent. Angioplasty (ballooning) crushes the hyperplastic tissue but does not correct the underlying flow instability. In fact, by restoring the lumen diameter, it may temporarily restore the high-velocity jet that drives the vibration,

resetting the cycle.

8. Clinical Implications and Future Directions

The recognition of high-frequency vibration as a distinct pathogen in vascular access demands a shift in clinical and engineering strategies.

8.1 Precision Surveillance

The era of the "physical exam" must give way to digital quantification.

- **Routine Spectral Monitoring:** Dialysis clinics should adopt digital stethoscopes to track HLPR trends. A rising HLPR (> 0.35) should trigger early intervention *before* the access clots.²⁴
- **Wearable Radar:** Low-cost radar patches could monitor VWM during dialysis sessions, providing a "vibration exposure" metric analogous to radiation dosimetry.³¹

8.2 Engineering the "Quiet" Fistula

Future surgical and device interventions should prioritize hemodynamic stability.

- **Geometry Optimization:** Computational planning could identify patient-specific anastomotic angles that minimize Dean vortex shedding and flow separation.
- **Damping Grafts:** Synthetic grafts could be engineered with viscoelastic properties specifically tuned to absorb energy in the 40–300 Hz band, acting as a "shock absorber" for the downstream vein.
- **External Supports:** Devices like the VasQ¹ that constrain the anastomotic angle may function, in part, by stabilizing the flow and reducing the generation of high-frequency energy.

8.3 Conclusion

The human body is a resonant structure, and the arteriovenous fistula represents a unique bio-mechanical experiment in which the vascular system is pushed to its vibratory limits. The evidence—spanning from the mathematical matrices of TurtleFSI to the molecular signaling of *lncRNA MEG3*—converges on a single conclusion: the "bruit" is not merely a sound; it is a signal of cellular distress. By decoding the frequency spectrum of the human body, we unlock new pathways to preserve the vascular lifelines essential for millions of patients worldwide.

9. Technical Appendix: Data Tables

Table 2: The Biological Frequency Spectrum

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System	Structure	Frequency Range	Function/Effect
Vascular (Systemic)	Heart/Pulse	1 – 2 Hz	Perfusion, Cyclic Strain
Vascular (Pathological)	AVF Wall	40 – 700+ Hz	Stenosis, Intimal Hyperplasia
Cutaneous (Tactile)	Meissner Corpuscle	40 – 60 Hz	Detection of Flutter/Slip
Cutaneous (Tactile)	Pacinian Corpuscle	200 – 300 Hz	Detection of Fine Vibration
Skeletal	Osteoblast	10 – 50 Hz	Osteogenesis (Bone Density)
Muscular	Myocyte	30 – 60 Hz	Hypertrophy (Atrophy suppression)

Table 3: High-Low Peak Ratio (HLPR) Diagnostics

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HLPR Value	Flow Condition	Hemodynamic Stress	Clinical Prognosis
< 0.35	Laminar / Stable	Low OSI	Stable Remodeling

> 0.35	Transitional / Vortex Shedding	High OSI (> 0.1)	Progressive Stenosis
0.67 (Median)	Highly Turbulent	Extended High OSI	Critical Remodeling

Table 4: TurtleFSI Simulation Parameters for AVF

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Parameter	Value / Method	Rationale
Solver Type	Monolithic (FEniCS)	Solves fluid/solid simultaneously to prevent added-mass instability.
Time Step (Δt)	1.5×10^{-5} s	Required to resolve kHz-range fluctuations and ensure ALE stability.
Mesh Density	150k – 3M+ Elements	Resolution of Kolmogorov turbulence scales.
Element Order	P2-P1 (Taylor-Hood)	Second-order accuracy for velocity and pressure fields.
Coupling	Strong (Implicit)	Ensures exact conservation of momentum at the FSI interface.

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